INCH-POUND

MIL-DTL-17361/13B(SH)
19 September 2006

SUPERSEDING
MIL-C-17361/13A(SH)
13 March 1995

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPES AQB-A1600, AQB-A1604, NQB-A1600, AND NQB-A1604

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17361.

LIST OF TABLES

TABLE I. Requirements. 2
TABLE II. Continuous current settings. 2
TABLE III. Adjustable instantaneous trip settings amperes/maximum time (all 3 poles). 3
TABLE IV. Adjustable short-time delay pickup (STDPU) settings in amperes (STD) (all 3 poles) (for trip unit types AQB-A1604 only). 3
TABLE V. Adjustable short-time delay (STD) settings in seconds (for trip unit types AQB-A1604 only). 4
TABLE VI. Tripping characteristics – long-time delay (LTD) (all 3 poles). 4
TABLE VII. Test requirements. 4

LIST OF FIGURES

FIGURE 1. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 outline dimensions. 5
FIGURE 2. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 outline and mounting dimensions. 6
FIGURE 3. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 outline and mounting base dimensions. 7
FIGURE 4. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 typical wiring diagram, handle locking device, and mounting dimensions. 8
FIGURE 5. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 outline dimensions. 9
FIGURE 6. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQBA1604 outline dimensions details. 10
FIGURE 7. Circuit breaker type AQB-A1604 time-current response. 11
FIGURE 8. Standard mounting fixtures for shock testing of types AQB-A1600 and AQB-A1604 circuit breakers. 12

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TABLE I. Requirements.

Maximum rated continuous current	1600 amperes
Voltage and frequency	500 VAC, 60 Hz
Interrupting rating	75,000 amperes, 60 Hz, AC
Trip unit	Interchangeable
Current limiting fuses	Not applicable
Type of connections	Back (removable mtg.)
Electric operator	Available
Undervoltage trip or shunt trip	Either available
Auxiliary switch	Four (a) contacts; four (b) contacts
Handle lock	Not available
Handle boot	Not available

NOTE: NQB circuit breakers shall consist of an AQB type without trip unit and a modification kit of metallic parts.

TABLE II. Continuous current settings.

AQB-A1600 trip units only						
Element rating /1						
600Y	600YM	800Y	1000Y	1200Y	1400Y	1600Y
AQB-A1604 trip units only						
Continuous current setting (amperes)						
600	800	1000	1200	1400	1600	

NOTE:

/1 Identification marking of trip units:

Numerals indicate continuous current rating of the specific trip unit.
Letter “Y” denotes that the trip units are applicable to AQB-A1600.
Letter “M” denotes calibration for applications having a high inrush current.

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TABLE III. Adjustable instantaneous trip settings amperes/maximum time (all 3 poles).

AQB-A1600 trip units only /1						
Element rating	**LO /3**	**2**	**3**	**4**	**HI /3**	**Seconds /2**
	AC	**AC**	**AC**	**AC**	**AC**	**60 Hz AC**
600Y	2000	3000	4000	5000	6000	0.032
600YM	4000	6000	8000	10000	12000	
800Y	2000	3000	4000	5000	6000	
1000Y	2000	3000	4000	5000	6000	
1200Y	2000	3000	4000	5000	6000	
1400Y	4000	6000	8000	1000	12000	
1600Y	4000	6000	8000	10000	12000	

AQB-A1604 trip units only /1, /4								
Frame rating	**3X**	**4X**	**5X**	**6X**	**8X**	**10X**	**12X**	**15X**
1600	4800	6400	8000	9600	12800	16000	19200	24000

NOTES:

/1 The tolerance for each ampere setting shall not exceed ±10%.

/2 Time shall be measured at three times pickup current of instantaneous setting. For qualification tests only, not for routine calibration.

/3 The LO and HI settings only are calibrated. Others are approximate.

/4 Opening time shall be 0.025 seconds or less when tested at the 3X instantaneous pickup (for qualification tests only, not for routine calibration).

TABLE IV. Adjustable short-time delay pickup (STDPU) settings in amperes (STD) (all 3 poles) (for trip unit types AQB-A1604 only).

Continuous current setting	**2X**	**3X**	**4X**	**6X**	**8X**	**10X**
600	1200	1800	2400	3600	4800	6000
800	1600	2400	3200	4800	6400	8000
1000	2000	3000	4000	6000	8000	10000
1200	2400	3600	4800	7200	9600	12000
1400	2800	4200	5600	8400	11200	14000
1600	3200	4800	6400	9600	12800	16000

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TABLE V. Adjustable short-time delay (STD) settings in seconds (for trip unit types AQB-A1604 only).

STD "delay setting"		
1	**2**	**3**
0.025–0.055 /1	0.065–0.095	0.105–0.135

NOTE:

/1 Time delay with 2X STDPU settings shall be 0.020–0.065 seconds.

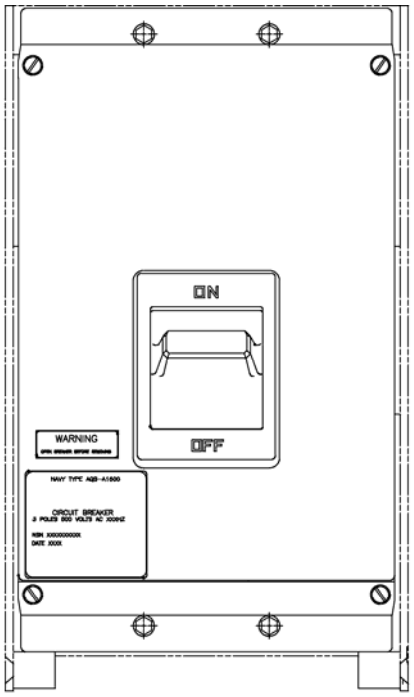
TABLE VI. Tripping characteristics – long-time delay (LTD) (all 3 poles).

AQB-A1600 trip units only	
Current as multiple of trip element rating	**Tripping time 60-Hz AC trip element**
1.50	Not less than 1 hour
2.25	Less than 1 hour
6.00	Between 20 and 32 seconds
AQB-A1604 trip units only	
Current as multiple of CCS	**Tripping time 60 Hz and 400 Hz AC**
1.35	Not less than 1 hour
1.65	Less than 1 hour
4.0	Between 35 and 55 seconds
6.0	Between 20 and 32 seconds

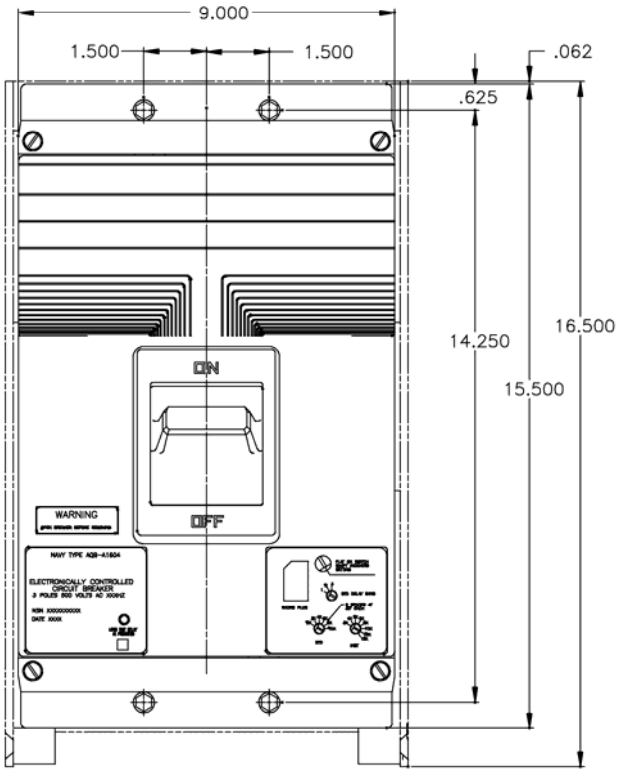
TABLE VII. Test requirements.

Mechanical endurance test	Mechanical (no-load) operations may be conducted between groups of load operations. The circuit breaker shall be subjected to 1000 close-open operations.		
Electrical endurance test	The test may be conducted in groups of 100 load operations. The rate of operation shall be 4 operations per minute for a total of 4000 close-open operations.		
Temperature rise tests	Size of cable and bus bars for temperature rise tests		
	Copper rating-amperes	Cable size	Copper bus size (inch)
	1600	Four (4) 400 MCM	Two (2) 3 x 1/4
Shock test	See Figure 2.		

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FRONT VIEW OF AQB–A1600 type CIRCUIT BREAKER



FRONT VIEW OF AQB–A1604 type CIRCUIT BREAKER
DIMENSIONS TYPICAL

FIGURE 1. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 outline dimensions.

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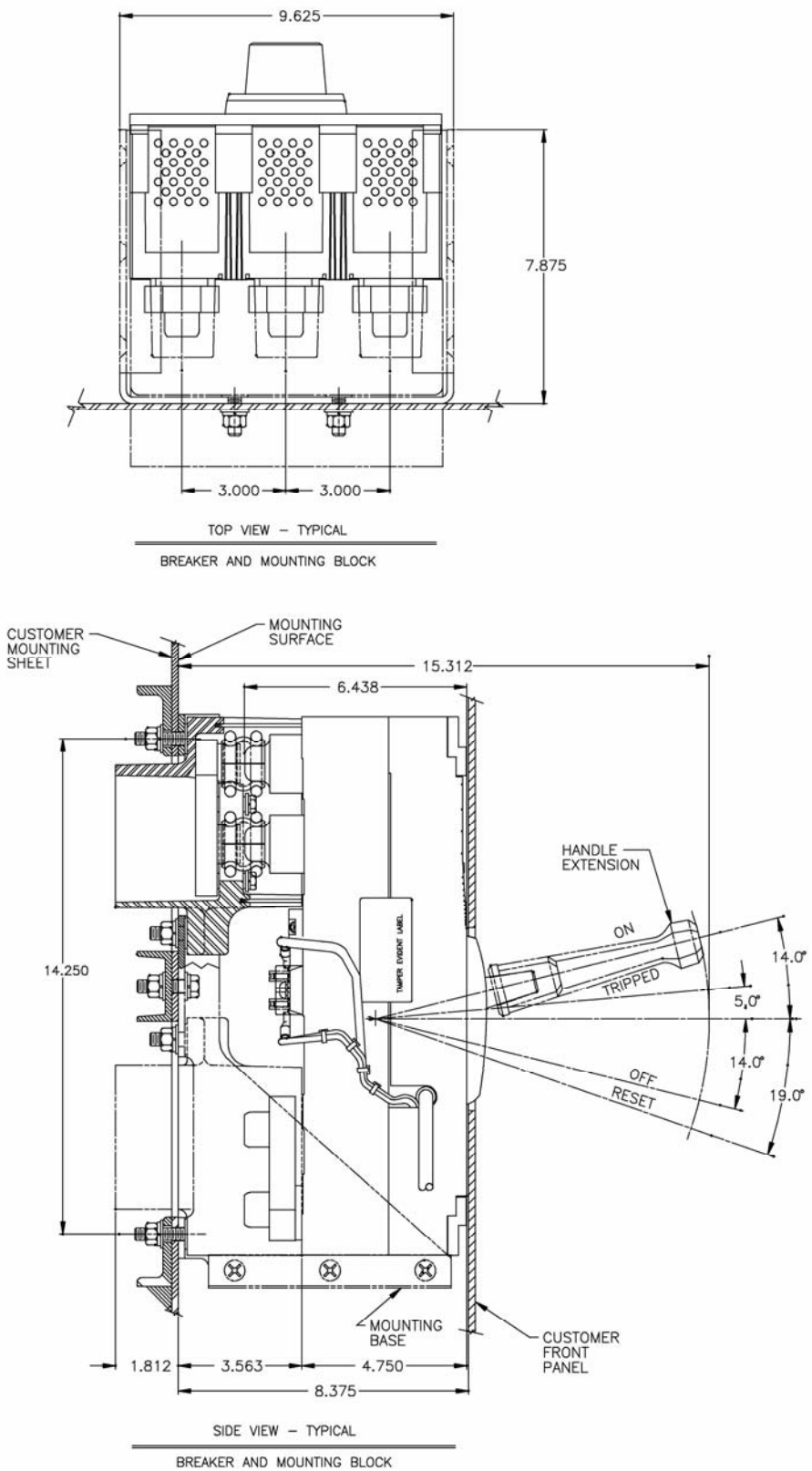


FIGURE 2. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 outline and mounting dimensions.

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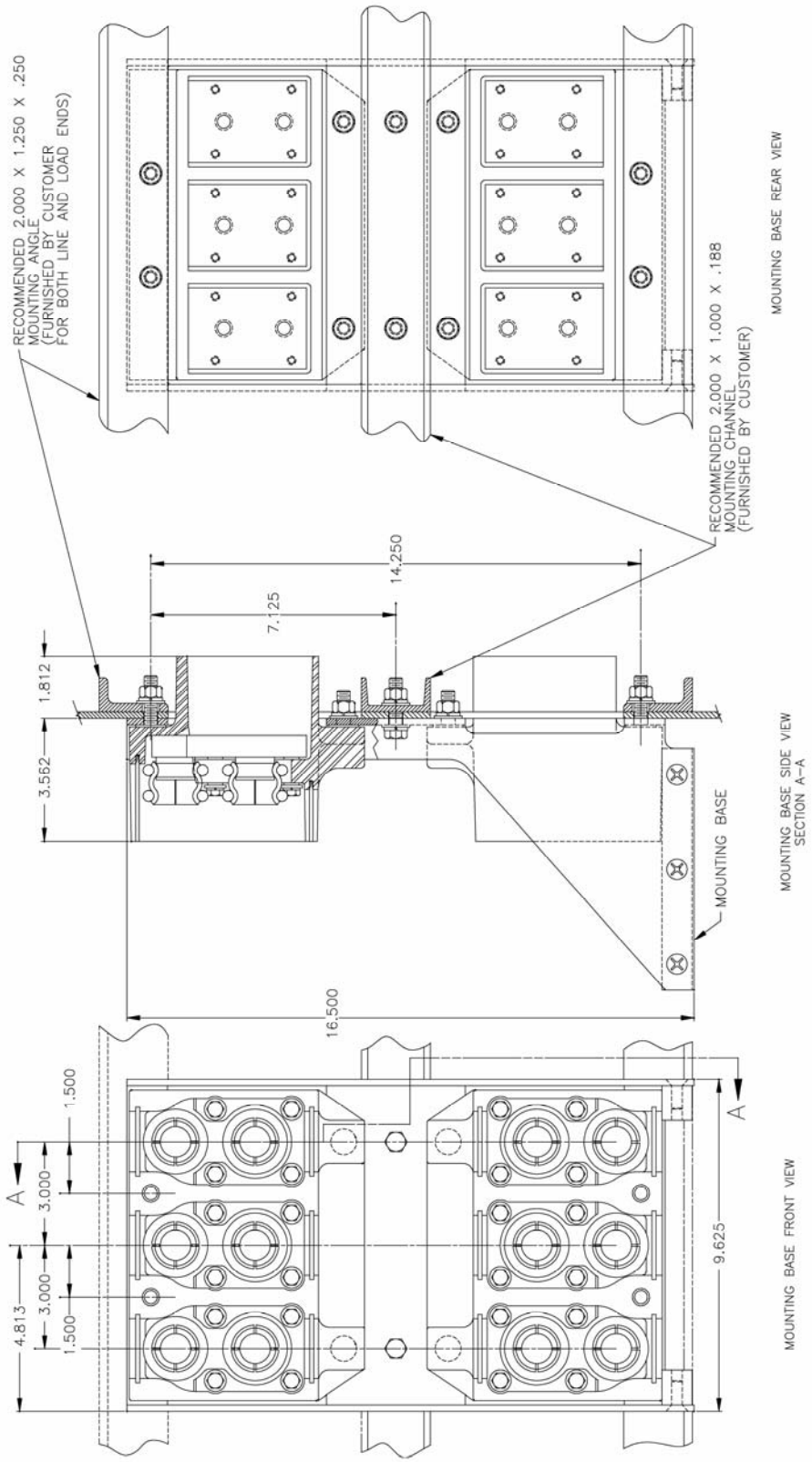


FIGURE 3. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 outline and mounting base dimensions.

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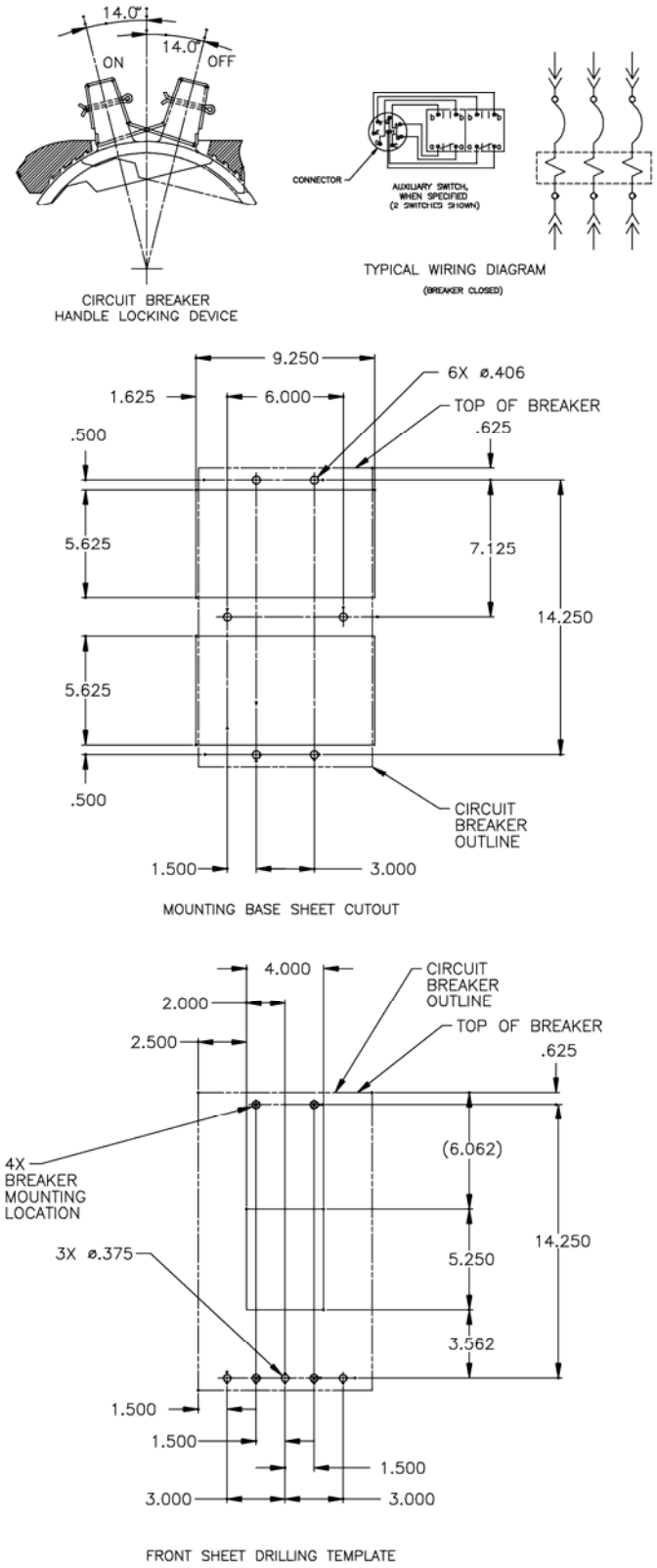


FIGURE 4. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 typical wiring diagram, handle locking device, and mounting dimensions.

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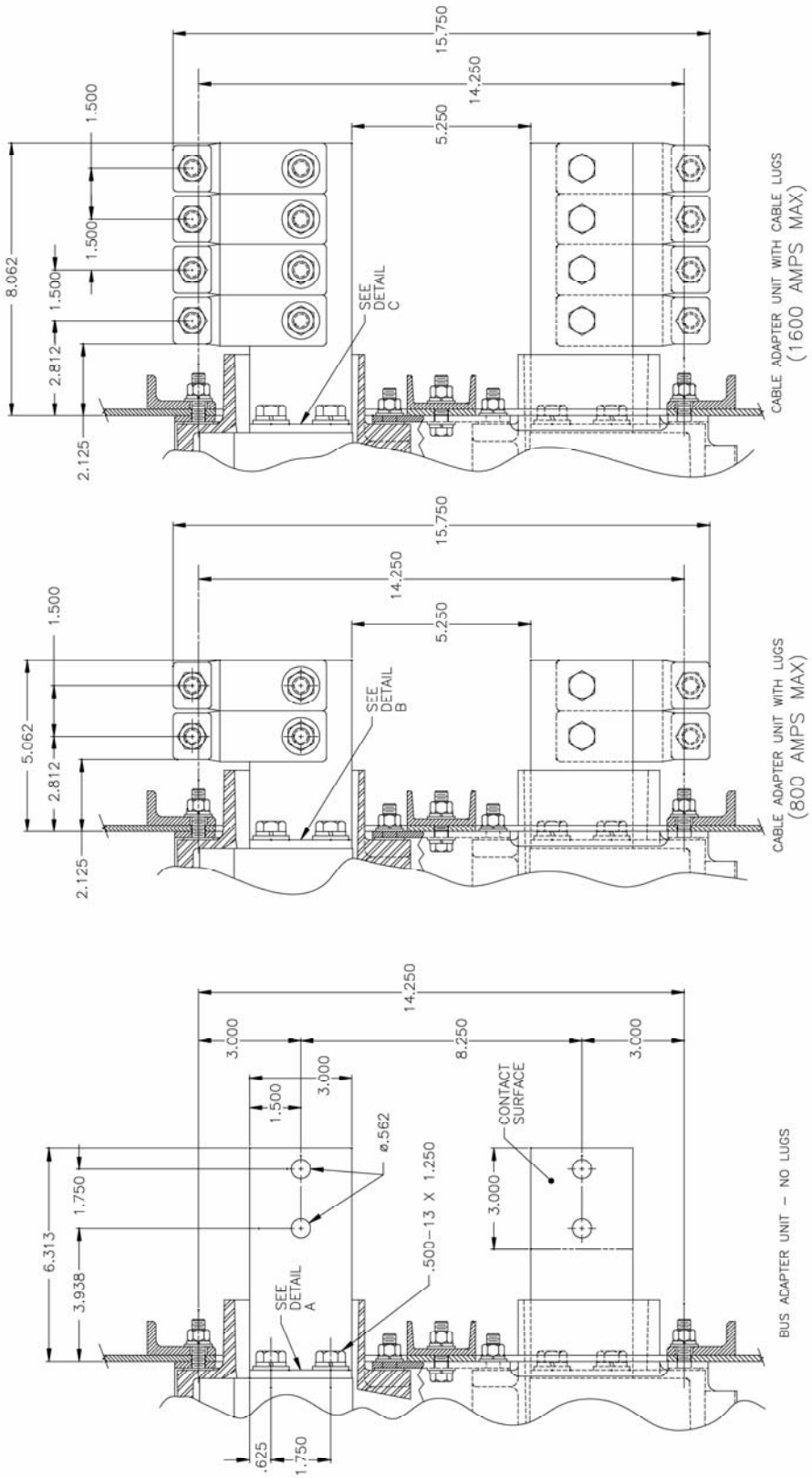


FIGURE 5. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQB-A1604 outline dimensions.

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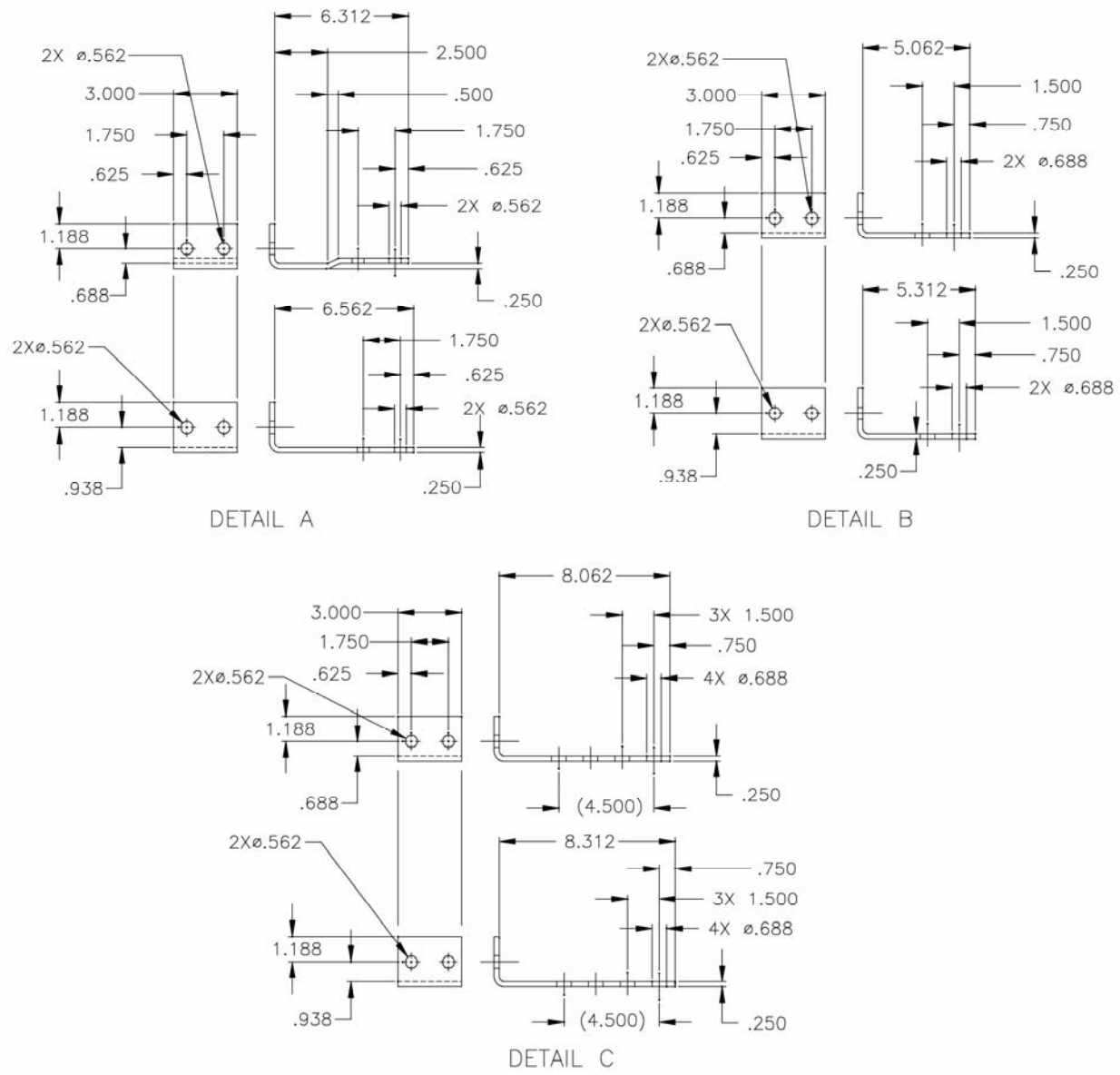


FIGURE 6. Circuit breaker types AQB-A1600, AQB-A1604, NQB-A1600, and NQBA1604 outline dimensions details.

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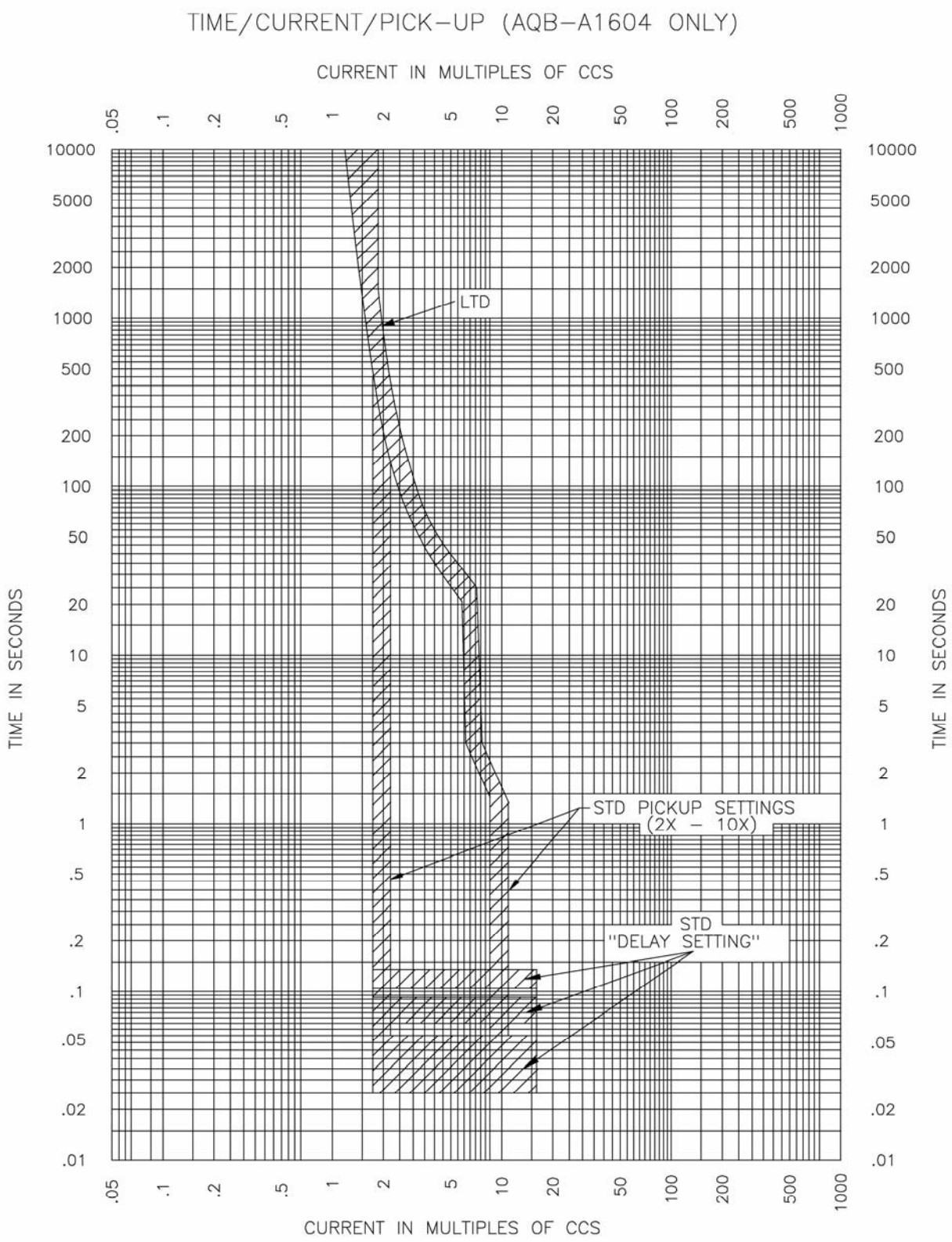


FIGURE 7. Circuit breaker type AQB-A1604 time-current response.

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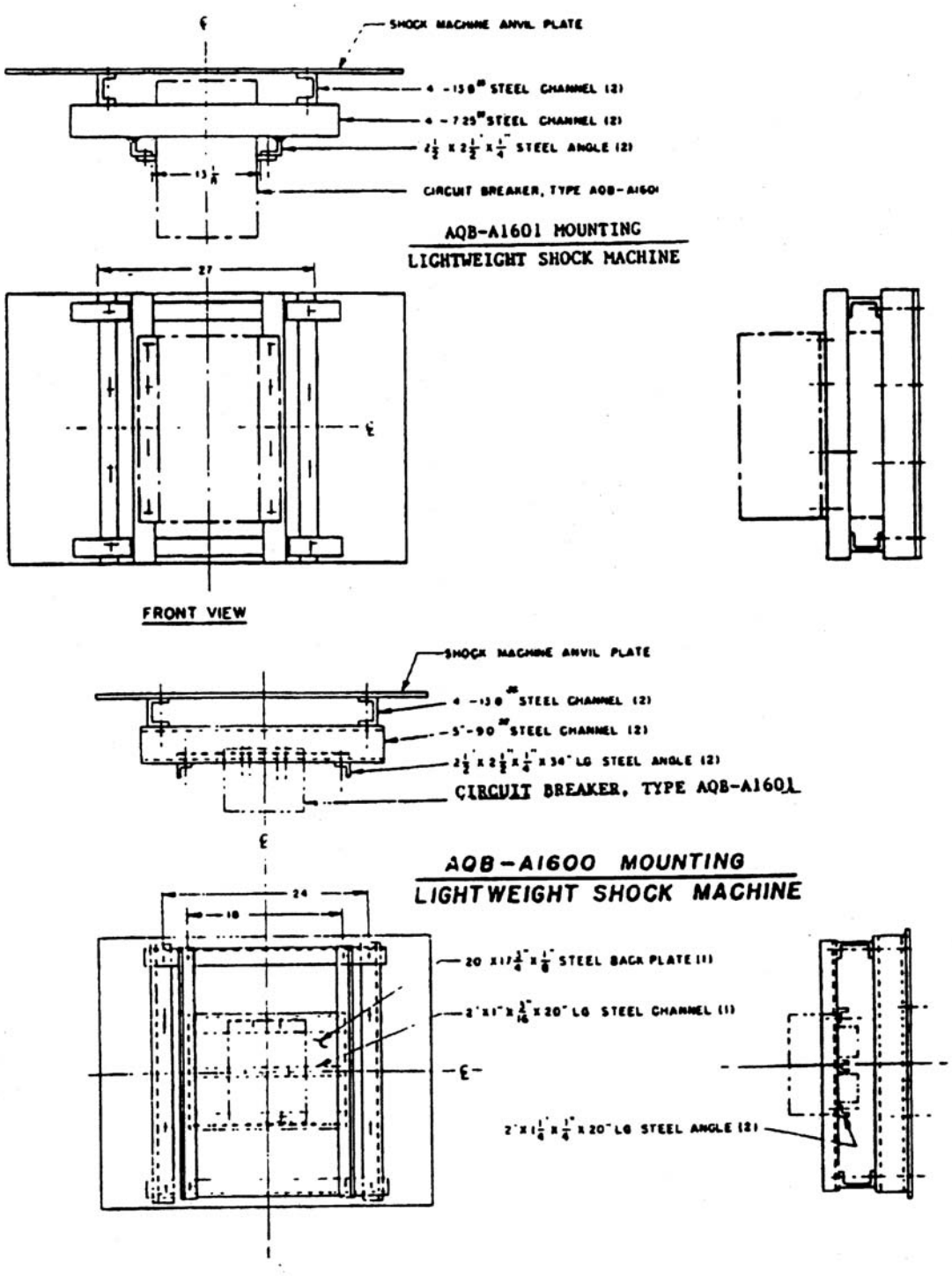


FIGURE 8. Standard mounting fixtures for shock testing of types AQB-A1600 and AQB-A1604 circuit breakers.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
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